

BSDA4001

DS/AI LAB COURSE · MILESTONE 1

AI-Powered Driver Wellness and Safety Monitoring System

Beyond drowsiness detection — a video-based, multi-model driver safety system with temporal deep learning, object-level compliance and trip intelligence.

Kushagra

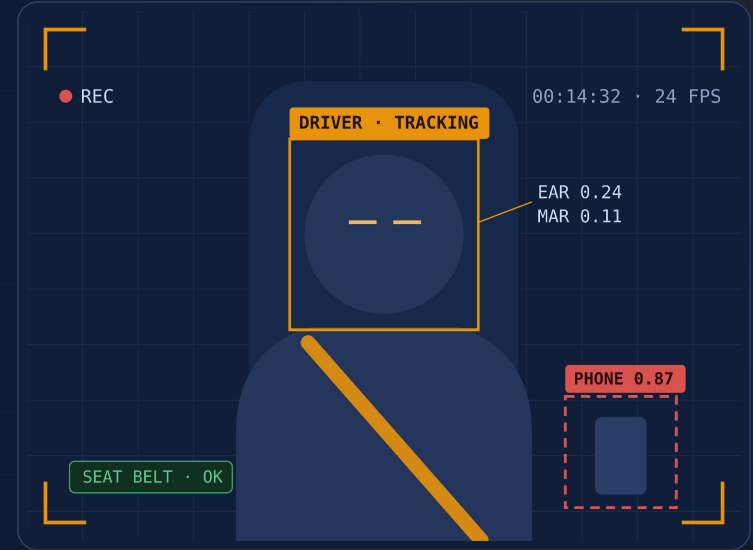
Shiwani

Sohini

Shubham

Ravina

Revised project direction after Milestone 1



LIVE MONITORING CONCEPT — dash-camera view with real-time detections

One System for the Whole Driver State

- Monitors overall driver state from continuous video
- Moves beyond a basic drowsiness detector
- Detects fatigue, distraction, seat belt, phone, smoking, drinking
- Tracks head pose and gaze-away behavior
- Fuses module outputs into one driver wellness score
- Generates trip-level and weekly/monthly driver reports

16–64

FRAME SEQUENCE WINDOWS

10

SYSTEM OUTPUTS

3

RISK LEVELS



Single Frames Cannot Explain Fatigue

Goal: real-time, video-based monitoring of fatigue, distraction, unsafe behavior and safety compliance.

WHY THE OLD APPROACH FAILS

- ✗ Single frames: a blink and drowsiness look identical
- ✗ Static frame counts and hardcoded thresholds
- ✗ Talking, singing or laughing trigger false alerts
- ✗ Frame-level YOLO has no sense of time

THE REVISED SYSTEM

- ✓ Learns fatigue as it develops across video sequences
- ✓ CNN-LSTM, CNN-GRU, TCN or lightweight 3D CNN
- ✓ MediaPipe/OpenCV: auxiliary features, not decision logic
- ✓ YOLO reserved for object-level safety tasks
- ✓ Fusion layer → Safe / Caution / High Risk

Why This Project, Why Now



01

Road safety need

- Fatigue drives crashes on night and long-distance routes
- Drowsiness is not the only unsafe state
- Phone use, missing seat belt, smoking, gaze-away



02

Course learning goals

- Full ML lifecycle, not plug-and-play detection
- Dataset collection, EDA, video preprocessing
- Sequences, training, evaluation, local deployment



03

Practical relevance

- Fleets monitor wellness and risky behavior
- Cab agencies and logistics: safer operations
- Drivers get early warnings; reports reveal trends

What We Build — and Deliberately Don't

IN SCOPE

- ✓ Video-based fatigue detection with temporal models
- ✓ Landmark temporal features via MediaPipe/OpenCV
- ✓ Distraction and driver activity classification
- ✓ YOLO: seat belt, phone, smoking, drinking
- ✓ Wellness score, event logs, trip reports
- ✓ Local dashboard demo + full dataset lifecycle

OUT OF SCOPE

- ✗ Real-car braking, steering or vehicle control
- ✗ Certified commercial safety product
- ✗ Production hardware or cloud fleet platform
- ✗ Full mobile application development
- ✗ Medical diagnosis of the driver
- ✗ ECG, EEG or other physiological sensors

Academic prototype — video and computer-vision based, not a certified safety system

Who Benefits From Safer Drivers



Drivers

Warnings for fatigue, distraction, seat belt, unsafe behavior



Passengers

Safer rides through earlier driver warnings



Cab agencies

Safety patterns during rides and night operations



Logistics & trucking

Fatigue tracking on long-distance transport



Bus operators

Passenger safety on long and night routes



Fleet owners

Wellness scores reveal risky driving trends



Safety reviewers

Logs expose repeated driver safety issues

DIRECT USERS

drivers & vehicle operators

INDIRECT

passengers, fleets, road-safety stakeholders

Ten Objectives, Three Phases

1

PHASE 01 — DATA FOUNDATION

- 1 Collect fatigue, distraction and safety datasets
- 2 EDA: classes, balance, duration, lighting, quality
- 3 Preprocess: frames, crops, fixed-length sequences

2

PHASE 02 — MODEL DEVELOPMENT

- 4 Temporal fatigue: CNN-LSTM, CNN-GRU, TCN, 3D CNN
- 5 MediaPipe features: EAR, MAR, head pose, gaze
- 6 YOLO: seat belt, phone, smoking, drinking
- 7 Distraction CNN: MobileNetV3, EfficientNet-B0, ResNet18

3

PHASE 03 — SYSTEM & EVALUATION

- 8 Fuse outputs into a driver wellness score
- 9 LLM trip reports from structured logs
- 10 Metrics: accuracy, F1, mAP, FPS, false alarms

Datasets: Collection and Understanding

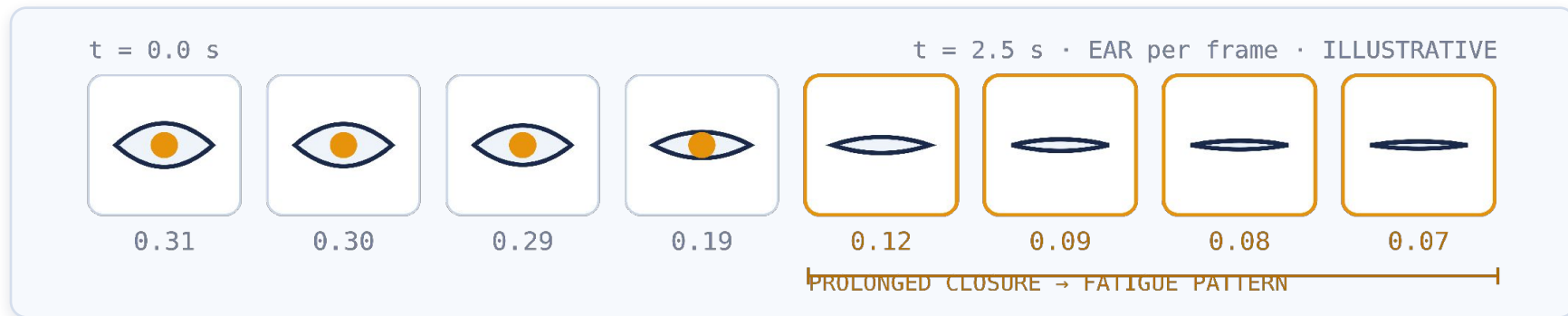
DATASET	TYPE	PURPOSE
NTHU Driver Drowsiness	Driver video	Fatigue and drowsiness patterns from video sequences
YawDD	Yawning video	Continuous yawning and mouth-movement patterns
State Farm Distracted Driver	Activity images	Texting, drinking, reaching and safe-driving classes
Drive&Act	Multi-modal video	Fine-grained driver action recognition
DMD Driver Monitoring	Video, multi-modal	Distraction, gaze and drowsiness scenarios
Kaggle / Roboflow sets	Object detection	Seat belt, phone, smoking, drinking detection

EDA class balance · duration · FPS · lighting · annotation quality · usable splits

PREP face/upper-body crops → normalized 16/32/64-frame sequences
→ splits

Final selection depends on access, license, size and Google Colab feasibility.

Video-Based Fatigue Detection



WHY TEMPORAL MODELS

- Blink vs drowsiness: identical in one frame
- Open mouth: yawn, talk, laugh or sing?
- Patterns: slow blinks, closures, yawns, nodding

CANDIDATE MODELS

CNN-LSTM — frame features + learned time behavior

CNN-GRU — similar, lighter and faster

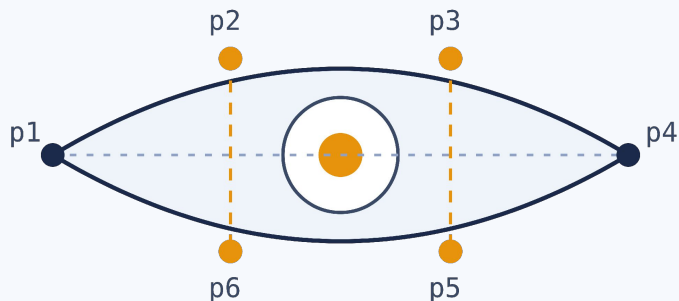
TCN — efficient temporal patterns

3D CNN (light) — spatial + temporal jointly

CNN + Attention — focuses on key frames

FIRST BUILD → **CNN-LSTM / CNN-GRU (Colab-feasible)**

Landmark-Based Temporal Features



$$\text{EAR} = (|p2-p6| + |p3-p5|) / 2 \cdot |p1-p4|$$

Eye Aspect Ratio – six landmarks per eye (MediaPipe / OpenCV)

EAR

MAR

PITCH

YAW

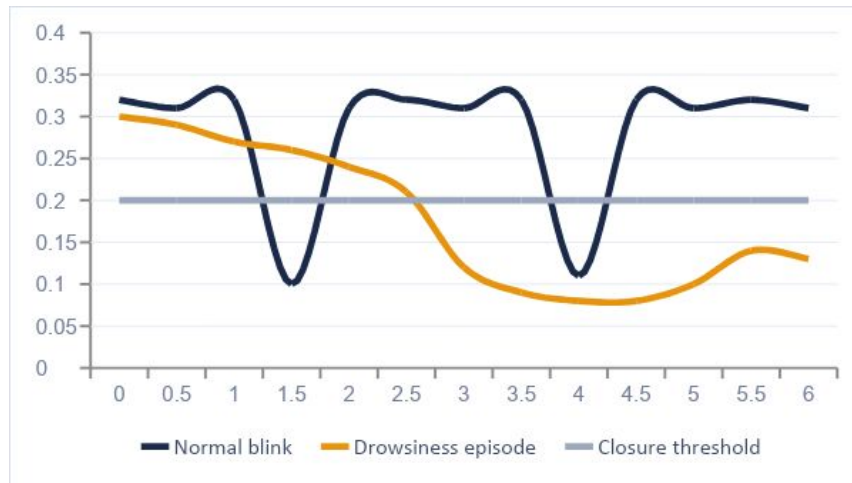
ROLL

BLINK

GAZE

extracted per frame → time-series input

EAR OVER TIME – ILLUSTRATIVE

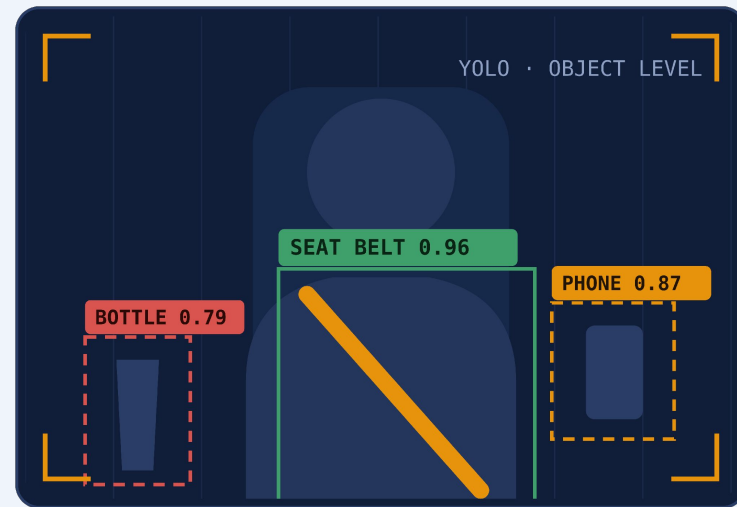


Sequences → LSTM / GRU / TCN. Learned patterns replace fixed thresholds; EAR/MAR/pose trends make every prediction explainable.

Distraction and Object-Level Safety

DISTRACTION CLASSIFIER — CNN

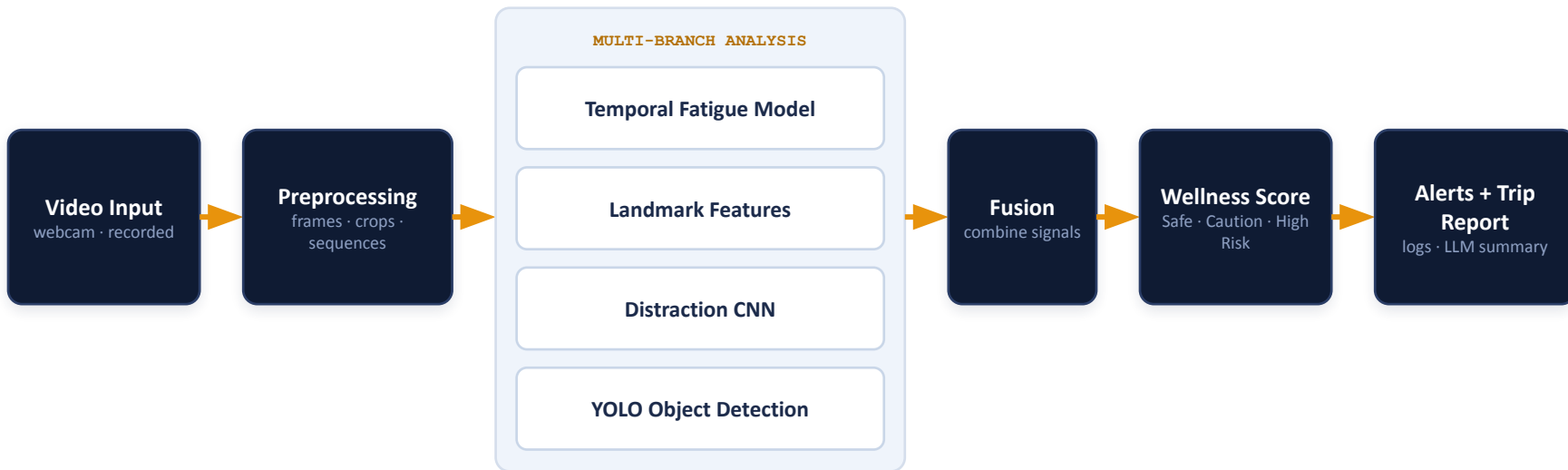
- Classes: texting, phone call, drinking, reaching, radio, grooming
- Plus a safe-driving baseline class
- Models: MobileNetV3, EfficientNet-B0, ResNet18
- CNN-LSTM upgrade if video/action data available
- Image baseline first, temporal upgrade later



Seat belt · phone · smoking · drinking — YOLOv8n / YOLO11n

Objects are frame-visible — YOLO fits there; fatigue stays video-sequence based

Proposed Architecture — Multi-Branch Pipeline



WHAT CHANGED FROM THE OLD DESIGN

Frame-level YOLO → **video-sequence temporal modeling**

Hardcoded thresholds → **learned fatigue patterns**

Drowsiness-only → **full wellness monitoring + reporting**

Wellness Score and Trip Reports

DRIVER WELLNESS SCORE

- Inputs: fatigue state, landmark trends, distraction class
- Plus seat belt, unsafe objects, gaze direction

 **Safe** — driver behavior appears normal

 **Caution** — mild fatigue or distraction detected

 **High Risk** — strong fatigue or multiple risk signals

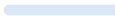
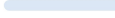
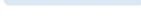
Shown on the dashboard · stored in trip logs

TRIP REPORTS — LLM LAYER

- LLM reads structured JSON/CSV logs — never raw video
- Llama 3.3 70B via API; smaller model for local demo
- Add-on layer — never replaces the ML models

TRIP REPORT

WEEKLY · MONTHLY

Trip summary	
Driver wellness score	
Safety risk summary	
Fatigue trend	
Distraction trend	
Seat belt compliance	
Safer-driving suggestions	

GENERATED BY LLM FROM STRUCTURED JSON/CSV LOGS

duration · events
violations ·
state
times → summary,
trends,
compliance,
suggestions

Evaluation Plan and Testing

METRICS BY MODULE

Fatigue

Accuracy, precision, recall, F1, confusion matrix, sequence-level accuracy

Priority

High recall — missed fatigue is dangerous; precision limits alert fatigue

Object detection

mAP, precision, recall, class-wise accuracy, inference FPS

Real-time

FPS, latency, inference time, local webcam demo performance

False alarms

Talking ≠ yawning · blink ≠ fatigue · mirror check ≠ distraction

TESTING SCENARIOS

Lighting

bright · low light · night-like ·
backlight

Driver

glasses · sunglasses · varied
camera angles

Behavior

blinks · closure · yawning · talking ·
laughing · look-away

Compliance

phone use · drinking · seat belt on
/ off

Tested on recorded videos and live local webcam

Team and References

Kushagra

Revised scope,
architecture,
documentation,
integration planning

Shiwani

Dataset collection,
dataset EDA, video data
preparation

Sohini

Landmark features,
metrics validation,
documentation review

Shubham

Deep learning model
design, temporal pipeline,
YOLO modules

Ravina

Evaluation plan, testing
scenarios, reporting
review

DATASETS

NTHU · YawDD · State Farm · Drive&Act · DMD

TOOLS

MediaPipe · OpenCV · Ultralytics YOLO

RESEARCH

CNN-LSTM / GRU, TCN, 3D CNN · EAR, MAR, PERCLOS, head
pose

REPORTING

Llama / open-source LLM documentation

From drowsiness detection to complete driver wellness monitoring.